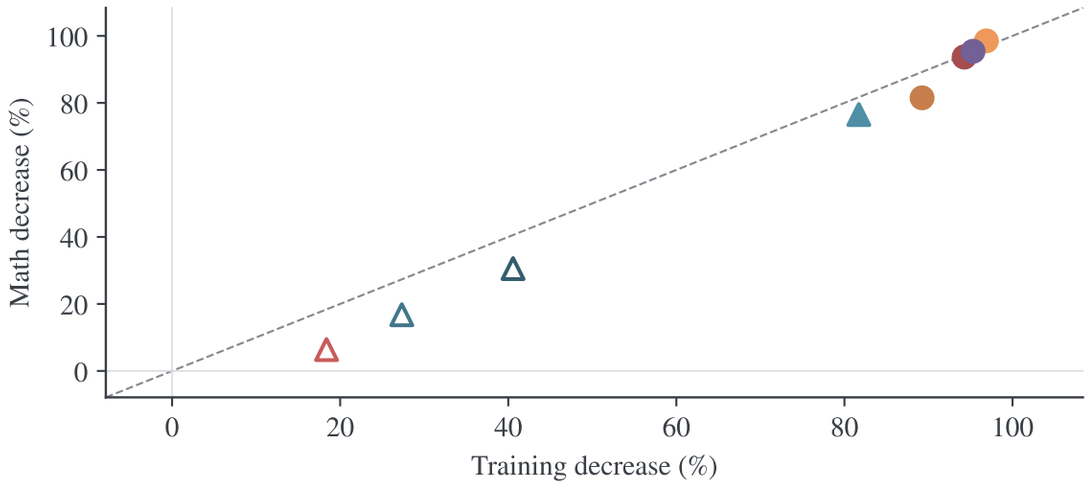
R1-1.5B | Training pool vs. held-out

Relative RKL decrease (%)

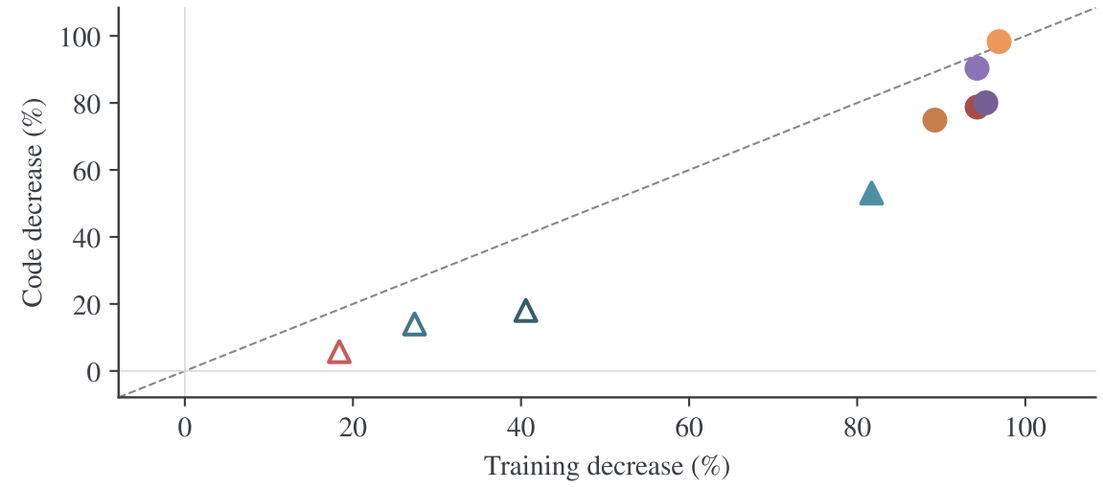
(a) Training vs. Math

n=9; r=1.00; rho=1.00



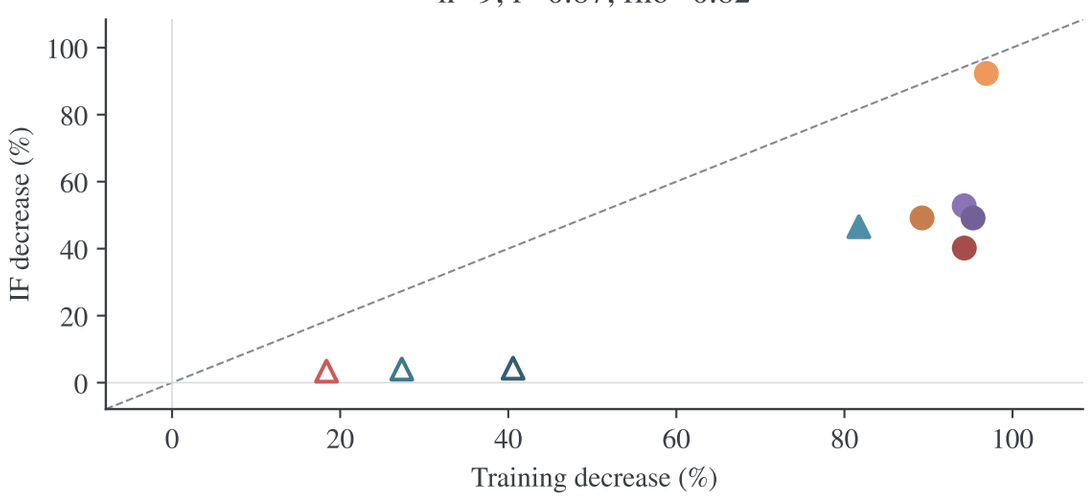
(b) Training vs. Code

n=9; r=0.97; rho=0.95



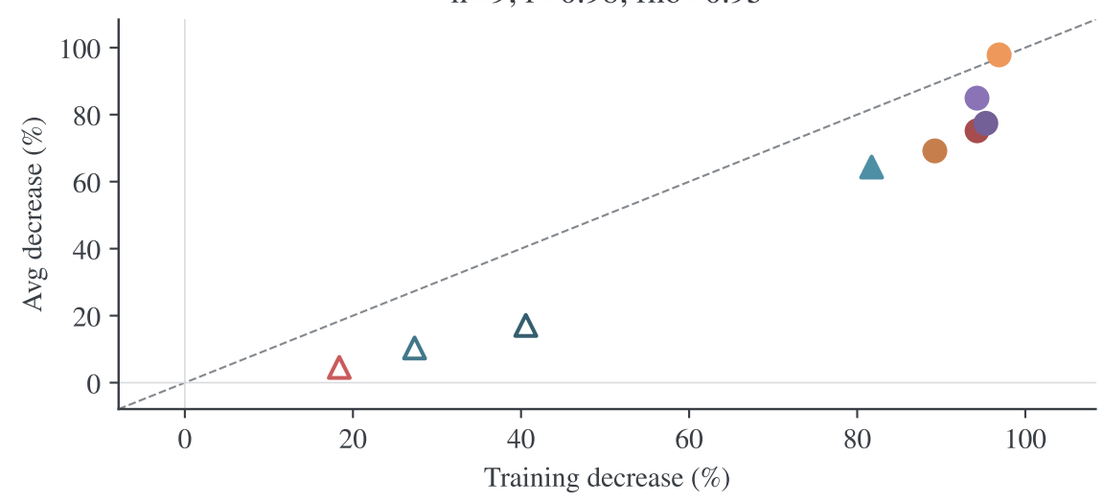
(c) Training vs. IF

n=9; r=0.87; rho=0.82



(d) Training vs. Avg

n=9; r=0.98; rho=0.95



DeepCoder-1.5B-Preview [64]

DeepScaleR-1.5B-Preview [320]

DeepSeek-R1-Distill-Qwen-7B [64]

JustRL-DeepSeek-1.5B [320]

Light-R1-7B-DS [64]

MiniMath-R1-1.5B [320]

Nemotron-Research-Reasoning-Qwen-1.5B / BroRL [320]

Nemotron-Research-Reasoning-Qwen-1.5B v1 [320]

Skywork-OR1-Math-7B [64]

Equal decrease

One point = one run endpoint; triangle: 64 updates; circle: 320; filled: same size; open: cross size.

Fixed-H RKL; dashed: equality; correlations pool all displayed endpoints (mixed budgets/sizes).